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Data Science

> Dealing with missing data
practical example

Missing values practical example

- **Exercise.....**
- Load the **managers_data** dataset
- Use **na.omit()** to store any full rows into **new_data** data frame
- Use **complete.cases()** to show all complete rows –store in **complete_data** and **missing_data** accordingly.
- Use **nrow()** to show a total complete rows
- Use **dim()** to show number of rows / cols in **complete_data**

Missing values practical example

- **Exercise.....**
- Use **na.omit()** to store any full rows into **new_data** data frame
- Use **complete.cases()** to show all complete rows –store in **complete_data** and **missing_data** accordingly.
- Use **nrow()** to show a total of all complete and missing rows

Missing values practical example

```
final_df <- na.omit(my_data)
dim(final_df)
final_df
```

```
complete_data <- my_data[complete.cases(managers),]
nrow(complete_data)
missing_data <- managers[!complete.cases(managers),]
nrow(missing_data)
missing_rows <- nrow(complete_data) - nrow(missing_data)
missing_rows
```

Identifying missing values

- The **md.pattern()** function in the **mice** package produces a tabulation of the missing data patterns in a matrix or data frame.
- Applying this function to our dataset produces the following

```
> md.pattern(managers)
```

	Date	Country	Gender	Q1	Q2	Q3	AgeCat	Age	Q4	Q5	Answer	total	mean	value
3	1	1	1	1	1	1	1	1	1	1	1	1	0	
1	1	1	1	1	1	1	1	1	0	0	0	0	4	
1	1	1	1	1	1	1	1	0	1	1	1	1	1	
	0	0	0	0	0	0	0	1	1	1	1	1	5	

Missing values
per row

Identifying missing values

- First row describes no missing rows eg 3 rows are ok
- 1 = no missing attribute, 0 = missing attribute
- Second row shows no missing data apart from Q4, A5 values
- Third row shows nothing missing apart from Age

Identifying missing values

- Age, Q4 and Q5, Answer total, mean value contain missing data
- 1 record has missing Age data, 1 has missing Q4 and Q5 data, Answer total and mean value data.

	Date	Country	Gender	Q1	Q2	Q3	AgeCat	Age	Q4	Q5	Answer total	mean value	
3													0
1													4
1													1
	0	0	0	0	0	0	0	1	1	1	1	1	5

Identifying missing values

- We can also show our missing data visually using the **VIM** package to access its **aggr()** function
- We use the **aggr()** function as follows:
- Identifying missing values

prop = proportions, numbers either shows or suppresses the numeric labels

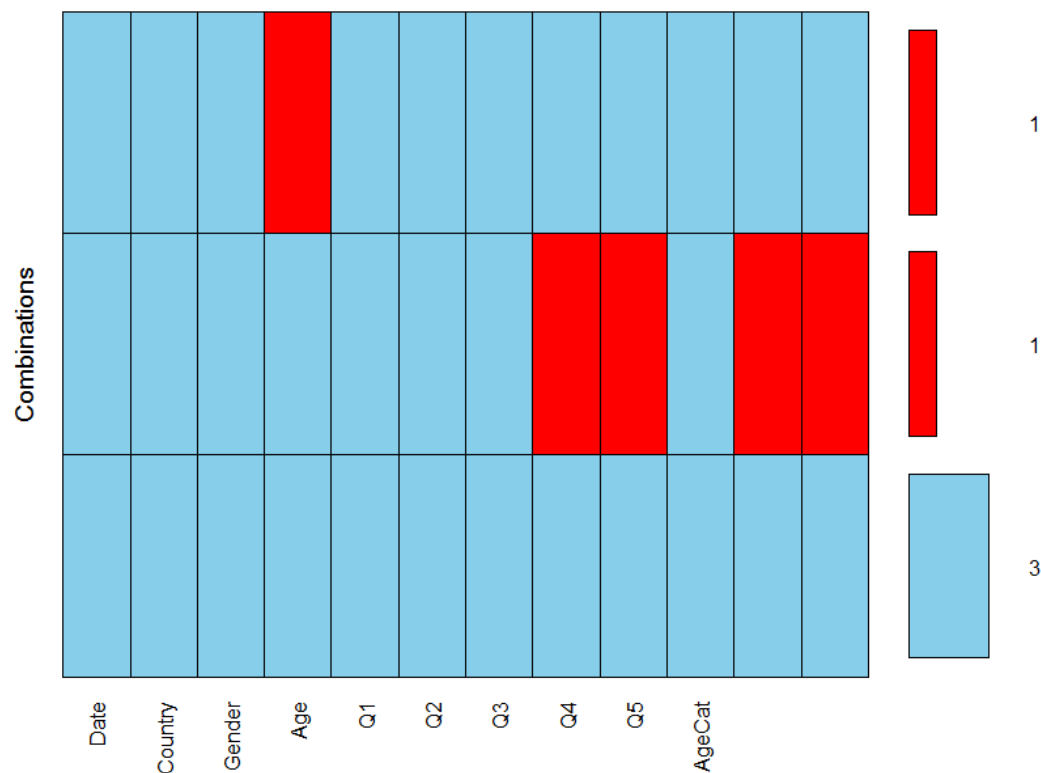
```
missing_values<-aggr(managers, prop = FALSE, numbers = TRUE)
```

Show summary of the contents of missing_values

```
summary(missing_values)
```

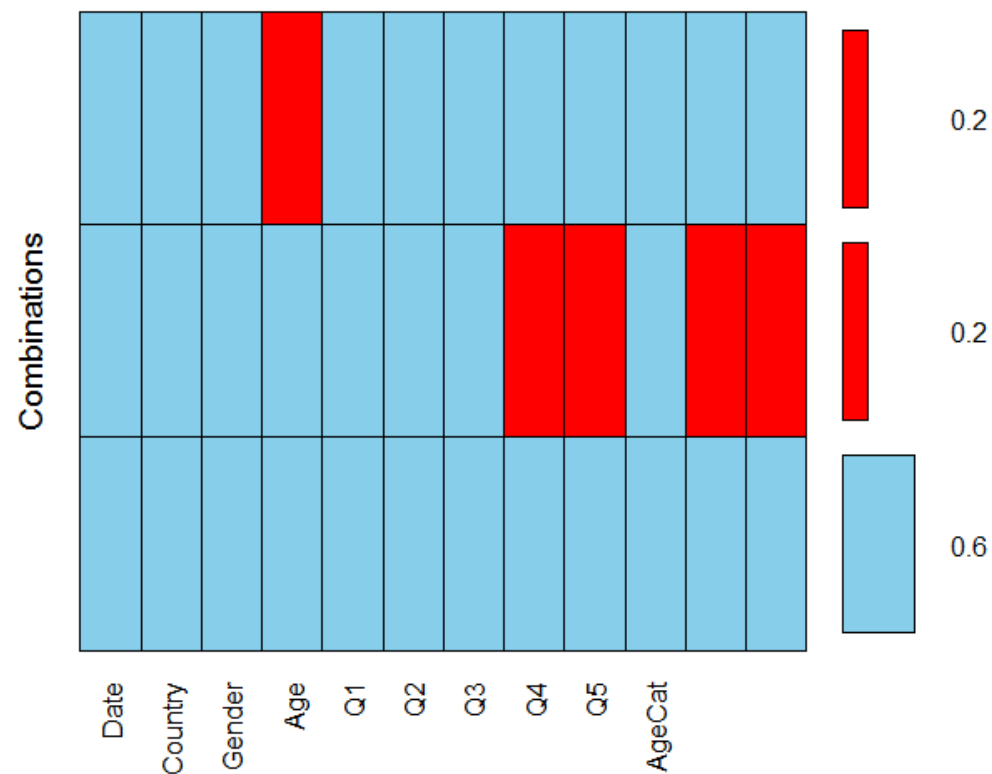
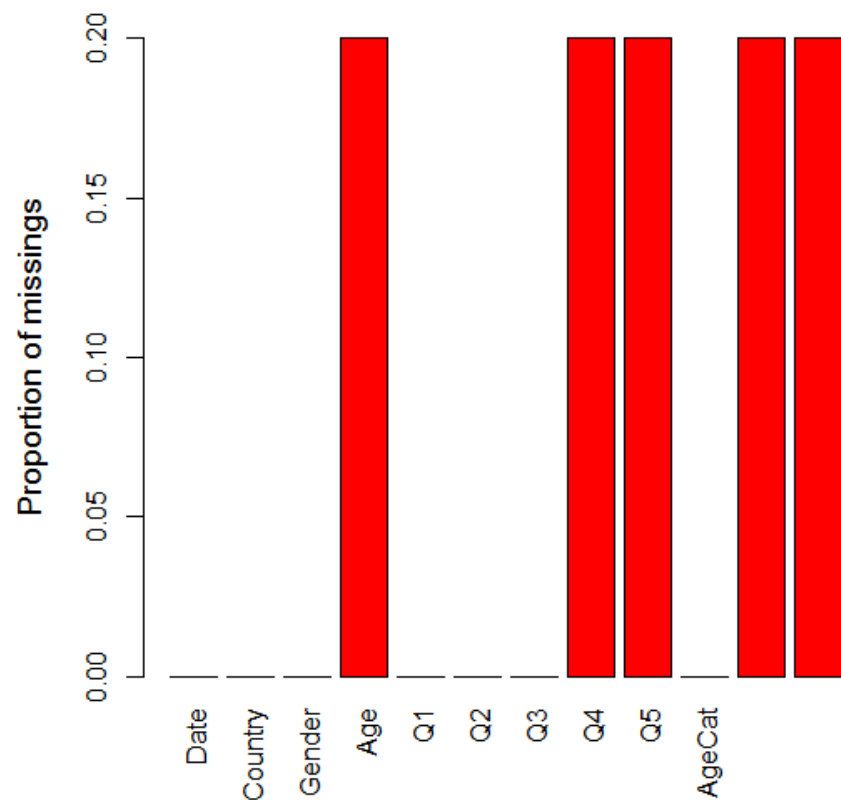
- # See this link for more info
<https://www.rdocumentation.org/packages/VIM/versions/4.8.0/topics/aggr>
-

- Age, Q4 and Q5, Answer total, mean value contain missing data



Identifying missing values

- Showing data proportionally results in this chart



Identifying missing values

- We can view a summary of the missing data

Missings per variable:

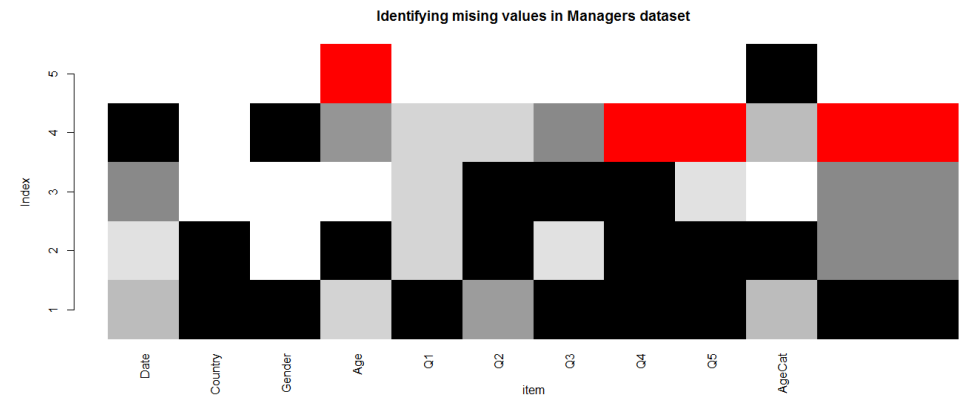
Variable	Count
Date	0
Country	0
Gender	0
Age	1
Q1	0
Q2	0
Q3	0
Q4	1
Q5	1
AgeCat	0
Answertotal	1
mean value	1

Missings in combinations of variables:

Combinations	Count	Percent
0:0:0:0:0:0:0:0:0:0:0:0	3	60
0:0:0:0:0:0:0:1:1:0:1:1	1	20
0:0:0:1:0:0:0:0:0:0:0:0	1	20

Identifying missing values - matrixplot

- The **matrixplot()** function produces a plot displaying the data for each case
- Data rescaled between range 0 –1 and represented in grayscale
- Missing values represented in red



Identifying missing values - correlations

- We can create a shadow matrix (recode data 1 = missing, 0 = present) for data
- Then present these as a correlation matrix to see if a relationship exists between attribute “missingness”.

```
# Show 1 = missing and 0 = okay for all data
# and convert to a data frame to create a correlation of missing data
missing_values<-as.data.frame(abs(is.na(my_data)))
head(missing_values)
# Examine each element in missing values and only
# store their output if value >0
# Used to create a correlation matrix between missing values
correlation_matrix<-missing_values[(apply(missing_values, 2, sum) > 0)]
correlation_matrix
# Show correlation matrix between extracted values
# 1 = perfect positive correlation, -1 = perfect negative correlation
# and 0 = no correlation (no link)
• cor(correlation_matrix)
```

Identifying missing values - correlations

- Resultant correlation matrix –small negative correlation between Age and Q4, Q5

```
> correlation_matrix<-  
missing_values[(apply(missing_values, 2, sum) > 0)]  
> correlation_matrix
```

```
Age Q4 Q5 summary_col  
1  0  0  0           0  
2  0  0  0           0  
3  0  0  0           0  
4  0  1  1           1  
5  1  0  0           0  
> cor(correlation_matrix)  
           Age      Q4      Q5 summary_col  
Age      1.00 -0.25 -0.25      -0.25  
Q4      -0.25  1.00  1.00       1.00  
Q5      -0.25  1.00  1.00       1.00  
summary_col -0.25  1.00  1.00       1.00
```

Identifying missing values

```
# removes any rows that contains NA -listwisedeletion
```

```
new_data<-na.omit(my_data)
```

```
new_data
```

```
# Use complete.casesto show rows where data is missing
```

```
missing_data<-complete.cases(my_data)
```

```
missing_data
```

```
# Show sum of missing rows
```

```
sum(missing_data)
```

```
# list the rows that do not have missing values
```

```
# Note that the ',' and no number inside square brackets means "all columns"
```

```
complete_data<-my_data[complete.cases(my_data),]
```

```
complete_data
```

Identifying missing values

```
# List rows with missing values
```

```
my_data[!complete.cases(my_data),]
```

```
# Find sum of all missing values in the age attribute
```

```
sum(is.na(my_data$Age))
```

```
sum(my_data[complete.cases(my_data)])
```

```
# Find the mean of missing values from the Age attribute
```

```
mean(is.na(my_data$Age))
```

```
# Find the mean of rows with no incomplete data
```

```
# Note that we dontneed to add the ',' as we're only
```

```
# looking for an overall mean of rows with missing values
```

```
mean(!complete.cases(my_data))
```

Interactive test using Swirl - missing values

- Run through the **5 -Missing Values** lesson in swirl.

```
library(swirl)
```

```
swirl()
```

- Follow the on-screen questions and complete the interactive test.
- Choose “**5 -Missing values**” as your lesson
- This will probably take no longer than 10 –15 minutes.